

adr_index

ADR Index

Field	Value
Revision	1
Created	2026-06-07
Last modified	2026-06-07
Status	active
Status summary	Master index of the five Helix OTA Architecture Decision Records (ADR-0001...ADR-0005), each with its one-line decision, status, and a relative link into <code>../research/adr/</code>. Summarizes the architectural through-line that ties the ADRs together. Derived directly from the five ADR files under <code>../research/adr/</code> and the master design (2026-06-07-helix-ota-design.md).
Issues	All five ADRs are still Proposed (operator review gate pending), so the recorded decisions are not yet ratified. ADR-0001's wrap target (hawkBit) is GATED on closing integration UNVERIFIEDs, with AOSP-native-only as the pre-approved fallback. Exact HelixConstitution clause numbering is UNVERIFIED (carried from the corpus convention).
Fixed	N/A (initial revision).
Continuation	Update each Status row in §3 as ADRs move Proposed → Accepted (or Superseded); when ADR-0001's gates resolve, record whether hawkBit is adopted or the Go-native fallback is taken; reconcile the cited §-clause numbers against the authoritative HelixConstitution once available (UNVERIFIED).

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The table-of-contents requirement is mandated by HelixConstitution §11.4.61 (UNVERIFIED clause number). This document carries its ToC immediately after the metadata table.

1. Purpose and scope

This document is the single entry point to the Helix OTA Architecture Decision Records. It lists each ADR with a one-line statement of the decision, the ADR's current status, and a relative link to the ADR source under `../research/adr/`, and it summarizes the **through-line** that connects the five decisions into one coherent architecture.

It is an index, not an authority: each linked ADR remains the controlling source for its own decision, drivers, options, consequences, and open items. Where this index restates a decision it does so in one line; the nuance (gates, fallbacks, sequencing) lives in the ADR. Per the corpus anti-bluff rule (§7.1 / §11.4.6 / §11.4.123, UNVERIFIED clause numbers), no fact is asserted here that is not present in the underlying ADRs, and unconfirmed items are tagged UNVERIFIED (see §6).

2. How to read this index

- **Decision (one line)** condenses the ADR's `Decision` section; it omits gating conditions and sequencing detail, which the ADR carries in full.
- **Status** is reproduced verbatim from each ADR's metadata. As of Revision 1 every ADR is **Proposed** (operator review gate pending); this index must be updated when an ADR is accepted or superseded (see the Continuation row).
- **Link** is a relative path from this file (`00-master/`) to the ADR under `research/adr/`, i.e. `../research/adr/<file>.md`.

3. ADR register

ADR	Decision (one line)	Status	Link
ADR-0001 — Wrapped engine	On device, AOSP update_engine (native A/B + Virtual A/B) is adopted/ wrapped under every option; for the server rollout/ campaign engine, wrap Eclipse hawkBit (front-	Proposed	../research/ adr/adr-0001- wrapped- engine.md

ADR	Decision (one line)	Status	Link
ADR-0002 — Supply-chain trust	runner) GATED on closing integration UNVERIFIEDs, with AOSP-native-only + a custom Go rollout engine as the pre-approved fallback; Mender rejected. Phased trust model: MVP (1.0.0) ships plain per-artifact trust (signed artifact + SHA-256 + AVB); TUF (go-tuf/v2) adopted server-side in 1.0.1+ with device-side enforcement gated behind an Android-client spike; the Uptane dual-repo model is a 1.0.1+ stretch goal; full Uptane and aktualizr rejected; signing interfaces designed MVP-forward.	Proposed	../research/adr/adr-0002-supply-chain-trust.md
ADR-0003 — Server topology	Adopt a modular monolith with extractable seams for 1.0.0-MVP — one Go binary with enforced internal module boundaries (rollout-engine and OS-adapter built as extractable modules); split into services only when a defined scale trigger fires.	Proposed	../research/adr/adr-0003-server-topology.md
ADR-0004 — Transport	Adopt the mandated transport: HTTP/3 (QUIC) primary with automatic HTTP/2 fallback (TLS 1.3), REST /api/v1 primary; two-class compression — Brotli→gzip for control-plane JSON, no content compression (ZIP_STORED, byte-identical) for OTA artifacts; device-pull streaming over HTTPS with mandatory HTTP Range + native resume.	Proposed	../research/adr/adr-0004-transport.md
ADR-0005 — Delta updates	MVP (1.0.0): full payload only (rely on Virtual A/B COW compression for on-device savings); post-MVP (1.0.1+): adopt AOSP-native incremental	Proposed	../research/adr/adr-0005-delta-updates.md

ADR	Decision (one line)	Status	Link
	payloads (ota_from_target_files -i) with full payloads retained as fallback; custom and third-party delta formats rejected; delta introduction gated on measured savings.		

4. The through-line

The five ADRs are not independent picks; they compose into one architecture whose spine is **native Android A/B on the device + a custom, decoupled Go control plane on the server**, with every wrap or hardening layer kept optional, gated, and behind a seam. Read in order:

1. **Device mechanism is fixed; only the server engine is in question (ADR-0001).** AOSP `update_engine` (native A/B + Virtual A/B, AVB/dm-verity, automatic boot-failure rollback) is adopted and wrapped — never replaced — on device under every option. The live choice is the *server* rollout/campaign engine: **wrap hawkBit (gated), else build it natively in Go** (pre-approved fallback). Either way the Go control plane stays the differentiator and the on-device path is unchanged.
2. **Trust is layered on top, server-side first (ADR-0002).** The MVP secures *which* artifact is authentic with plain per-artifact signing (SHA-256 + detached signature + AVB). A **TUF trust layer lands server-side now-ish (1.0.1+) and device-side at 1.0.1+** behind an Android-client spike; signing interfaces are designed MVP-forward so TUF/Uptane drop in without rework. Trust is complementary to delivery — `update_engine` still performs the atomic, anti-bricking apply.
3. **The Go control plane is a modular monolith (ADR-0003).** One deployable Go binary with enforced internal boundaries that mirror the future service seams (rollout-engine and OS-adaptor are the extractable first-movers). This keeps the MVP lean while preserving the decoupling that lets future OSes and future projects reuse the modules; services are carved out only when a concrete scale trigger fires. (If ADR-0001 adopts hawkBit, hawkBit remains a *separate* deployable behind the Go control plane, orthogonal to this topology.)
4. **Transport carries it over HTTP/3 + Brotli, with HTTP/2 fallback (ADR-0004).** Control-plane REST/JSON rides **HTTP/3 (QUIC) primary → HTTP/2 fallback**, with **Brotli→gzip** content compression; OTA artifacts ride the *same* edge but pinned to byte-identical ZIP_STORED with **HTTP Range** + native resume, so streaming verify-before-apply and signature checks hold. The two-class rule is what lets one transport serve both small JSON and large opaque payloads without breaking integrity.
5. **Payloads are full for MVP, delta later (ADR-0005).** MVP ships **full payload.bin only** (Virtual A/B COW compression covers on-device savings), with **AOSP-native incrementals adopted post-MVP (1.0.1+)** over the very same transport and the unchanged `applyPayload` path — additive, gated on measured savings, with full payloads kept as the universal fallback.

The common discipline across all five: **lock the device mechanism, keep the server custom and decoupled, defer complexity behind gates/seams, and never compromise verify-before-apply or byte-identity.**

5. Modules referenced by these ADRs

To honor catalogue-first and the anti-bluff rule, only **verified catalogue submodules** and the **six NEW ota-* submodules** are named in the corpus. The canonical catalogue list is in [documentation_standards.md §9](#); the full component→submodule binding is in [submodule_reuse_map.md](#). The decisions above are realized through:

- **Catalogue submodules** (reuse/extend), e.g. http3, middleware, ratelimiter, recovery, Storage, security, database, observability, Herald, config, cache, eventbus, discovery, mdns, containers, docs_chain/Document/Formatters; KMP: Auth-KMP, Security-KMP, Storage-KMP, Config-KMP. (Capabilities of specific submodules are UNVERIFIED pending API inspection — see the reuse map.)
- **The six NEW ota-* submodules** (no catalogue cover): ota-protocol, ota-artifact-validator, ota-rollout-engine, ota-update-engine-bridge, ota-android-agent, ota-telemetry-schema.

No other module names are introduced by this index.

6. Anti-bluff notes

Per HelixConstitution §7.1, §11.4.6, and §11.4.123 (UNVERIFIED clause numbers):

- **No fabricated facts.** Every decision one-liner and the through-line are condensed from the five linked ADRs' own Decision and Context sections; this index introduces no claim absent from those ADRs.
- **Status is reproduced, not inferred.** All five ADRs carry Status: Proposed in their metadata as of this revision; that is recorded verbatim in [§3](#) and must be re-synced when the ADRs change (see Continuation).
- **Gates and UNVERIFIEDs are not flattened.** ADR-0001's hawkBit choice is GATED with a Go fallback; ADR-0002's device-side TUF, ADR-0004's native download-resume default, and ADR-0005's COW/XOR savings figures are all marked UNVERIFIED in their source ADRs and must not be quoted as fact — consult the ADR before relying on any such figure.
- **Only real modules are referenced** — the verified catalogue submodules ([documentation_standards.md §9](#)) and the six NEW ota-* submodules ([submodule_reuse_map.md §4](#)); no module name is invented here.
- HelixConstitution clause numbers (§11.4.61, §7.1, §11.4.6, §11.4.123) are carried from the corpus convention and are UNVERIFIED against the authoritative constitution text.